Hannah Yeung (she/her)

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TECHNICAL SKILLS

Programming: Python, C++, C#, C, MATLAB, R
Engineering Tools: COMSOL, Simulink, AutoCAD, SolidWorks
Hardware & Electronics: Breadboarding, circuit prototyping, soldering, laser cutting, microcontroller integration, sensor interfacing
Laboratory Techniques: DNA extraction, gel electrophoresis, mammalian cell transfection, fluorescence microscopy imaging
Certifications: Biopharmaceutical Manufacturing, Standard First Aid & CPR, UBC Laboratory Safety

EDUCATION

University of California, San Diego — **September, 2025 – December, 2026**
Master of Engineering – Bioengineering

University of British Columbia — **September, 2021 – May, 2025**
Bachelor of Applied Science – Biomedical Engineering (With Distinction)

WORK EXPERIENCE

VisionLabs AI, Inc., San Francisco, CA — **January, 2026 - Present**
Automated Calibration Development Intern

- Develop and validate repeatable optical and geometric calibration procedures using precision XY rigs with micrometer-grade adjusters.
- Quantify and correct device-to-device optical variations across 20 units through controlled image capture and Python-based calibration analysis.
- Generate device-specific correction coefficient to improve measurement accuracy of an iPhone-based corneal topography system.
- Design calibration fixtures and visual targets and evaluate automation approaches using optical pattern recognition and machine vision.

Kumon Canada, Inc., Vancouver, BC — **July, 2018 – August, 2025**
Kumon Instructor

- Delivered personalized lessons tailored to individual learning capability to foster independent learning.
- Guided problem-solving using strategic hints to enhance comprehension.
- Graded assignments with efficiency and accuracy to ensure timely feedback.
- Assessed student progress regularly and communicated with parents to promote continuous improvement.

TECHNICAL PROJECTS

Robotics Surface Sampling System (Team Project), UBC — **March, 2025 - April, 2025**
- Modeled and simulated an automated surface-sampling robot in SolidWorks to reduce nosocomial infection risks in clinical environments through mechanical design, control, and human–robot interaction.

Organ Cryopreservation Chamber (Team Project), UBC — **September, 2024 - April, 2025**
- Designed and prototyped a subzero organ preservation system using thermoelectric cooling and Arduino-based temperature sensing, maintaining –1 °C for six hours to improve short-term tissue stability.

“Artificial Pancreas” System (Team Project), UBC — **January, 2024 – April, 2024**
- Implemented a PID-controller insulin infusion system in Simulink that reduced glucose overshoot from 2.46 to 2.27, enhancing system stability for Type 1 diabetes management.

CPR Monitoring System (Team Project), UBC — **January, 2024 - April, 2024**
- Built an Arduino-based CPR monitoring prototype incorporating haptic feedback and fNIR sensors to provide real-time assessment of cerebral perfusion and compression rate for out-of-hospital cardiac arrest patients.

INTERESTS & ACTIVITIES

- Swimming (Certified by The Lifesaving Society), Taekwondo (Black Belt), Languages (Cantonese & Mandarin)